Chapter 15, Problem 39P

Problem

Determine $f(t)$ if:

(a) $F(s) = \dfrac{2s^3 + 4s^2 + 1}{(s^2 + 2s + 17)(s^2 + 4s + 20)}$

(b) $F(s) = \dfrac{s^2 + 4}{(s^2 + 9)(s^2 + 6s + 3)}$

Step-by-step solution

Step 1 of 6

(a)

Write the function $F(s)$.

$$F(s) = \frac{2s^3 + 4s^2 + 1}{(s^2 + 2s + 17)(s^2 + 4s + 20)}$$

Write the function $F(s)$ into partial fractions.

$$F(s) = \frac{As + B}{(s^2 + 2s + 17)} + \frac{Cs + D}{(s^2 + 4s + 20)} \quad \text{...... (1)}$$

$$\frac{2s^3 + 4s^2 + 1}{(s^2 + 2s + 17)(s^2 + 4s + 20)} = \frac{As + B}{(s^2 + 2s + 17)} + \frac{Cs + D}{(s^2 + 4s + 20)}$$

Equate the numerators of the equation.

$$2s^3 + 4s^2 + 1 = (As + B)(s^2 + 4s + 20) + (Cs + D)(s^2 + 2s + 17)$$

Equate the coefficients of s^3, s^2, s.

$2 = A + C$ ...... (2)

$4 = 4A + B + 2C + D$ ...... (3)

$0 = 20A + 4B + 17C + 2D$ ...... (4)

Equate the constants on both sides of the equation (1).

$1 = 20B + 17D$ ...... (5)

Substitute $2 - A$ for C in equation (3).

$$4 = 4A + B + 2(2 - A) + D$$

$2A + B + D = 0$ ...... (6)

Step 2 of 6

Substitute $2-A$ for C in equation (4).

$$0=20A+4B+17(2-A)+2D$$

$$3A+4B+2D=-34 \quad \ldots\ldots (7)$$

Solve equations (5), (6) and (7).

$$A=-1.6$$
$$B=-17.8$$
$$D=21$$

Calculate the value of *C.*

$$\begin{aligned} C&=2-A \\ &=3.6 \end{aligned}$$

Substitute -1.6 for *A*, -17.8 for B, 3.6 for C, 21 for D in equation (1).

$$F(s)=\frac{-1.6s-17.8}{(s^2+2s+17)}+\frac{3.6s+21}{(s^2+4s+20)}$$

Step 3 of 6

Rewrite the function $F(s)$.

$$F(s)=\frac{-1.6s-17.8}{(s+1)^2+16}+\frac{3.6s+21}{(s+2)^2+16}$$

$$F(s)=\frac{-1.6s}{(s+1)^2+4^2}-\frac{17.8}{(s+1)^2+4^2}+\frac{3.6s}{(s+2)^2+4^2}+\frac{21}{(s+2)^2+4^2}$$

$$F(s)=\frac{-1.6(s+1)+1.6}{(s+1)^2+4^2}-\frac{17.8}{(s+1)^2+4^2}+\frac{3.6(s+2)-7.2}{(s+2)^2+4^2}+\frac{21}{(s+2)^2+4^2}$$

$$=\frac{-1.6(s+1)}{(s+1)^2+4^2}-\frac{\left(\frac{16.2}{4}\right)(4)}{(s+1)^2+4^2}+\frac{3.6(s+2)}{(s+2)^2+4^2}+\frac{\left(\frac{13.8}{4}\right)(4)}{(s+2)^2+4^2}$$

$$F(s)=\frac{-1.6(s+1)}{(s+1)^2+4^2}-\frac{(4.05)(4)}{(s+1)^2+4^2}+\frac{3.6(s+2)}{(s+2)^2+4^2}+\frac{(3.45)(4)}{(s+2)^2+4^2} \quad \ldots\ldots (8)$$

Step 4 of 6

Consider the Laplace transform pairs from Table 15.2 in the textbook.

$$e^{-at}\sin\omega t=\frac{\omega}{(s+a)^2+\omega^2}$$

$$e^{-at}\cos\omega t=\frac{s+a}{(s+a)^2+\omega^2}$$

Apply inverse Laplace transform to equation (8).

$$f(t)=\left(-1.6e^{-t}\cos 4t-4.05e^{-t}\sin 4t+3.6e^{-2t}\cos 4t+3.45e^{-2t}\sin 4t\right)u(t)$$

Therefore, the inverse Laplace transform of $F(s)$ is

$$\boxed{\left(-1.6e^{-t}\cos 4t-4.05e^{-t}\sin 4t+3.6e^{-2t}\cos 4t+3.45e^{-2t}\sin 4t\right)u(t)}.$$

Step 5 of 6

(b)

Write the function $F(s)$.

$$F(s)=\frac{s^2+4}{(s^2+9)(s^2+6s+3)}$$

Write the function $F(s)$ into partial fractions.

$$F(s)=\frac{As+B}{(s^2+9)}+\frac{Cs+D}{(s^2+6s+3)} \quad \text{...... (9)}$$

$$\frac{s^2+4}{(s^2+9)(s^2+6s+3)}=\frac{As+B}{(s^2+9)}+\frac{Cs+D}{(s^2+6s+3)}$$

Equate the numerators of the equation.

$$s^2+4=(As+B)(s^2+6s+3)+(Cs+D)(s^2+9)$$

Equate the coefficients of s^3, s^2, s.

$$0=A+C \quad \text{...... (10)}$$

$$1=6A+B+D \quad \text{...... (11)}$$

$$0=3A+6B+9C \quad \text{...... (12)}$$

Equate the constants on both sides of the equation (9).

$$4=3B+9D \quad \text{...... (13)}$$

Substitute $-A$ for C in equation (12).

$$0=3A+6B+9(-A)$$
$$A=B$$

Substitute B for *A* in equation (11).

$$1=6A+A+D$$
$$D=1-7A$$

Substitute B for *A* and $1-7A$ for D in equation (13).

$$4=3A+9(1-7A)$$
$$A=0.08333$$

Calculate the value of D.

$$D=1-7(0.08333)$$
$$=0.41669$$

Substitute 0.08333 for A and B -0.08333 for C, 0.41669 for D in equation (9).

$$F(s)=\frac{0.08333s+0.08333}{\left(s^2+9\right)}+\frac{-0.08333s+0.41669}{\left(s^2+6s+3\right)}$$

$$=\frac{0.08333(s+1)}{\left(s^2+9\right)}+\frac{-0.08333s+0.41669}{\left(s^2+6s+3\right)}$$

$$=\left(\begin{array}{l} 0.08333\frac{s}{s^2+9}+\left(\frac{0.08333}{3}\right)\frac{3}{s^2+9}-\frac{0.08333(s+3)-0.24999}{(s+3)^2-\left(\sqrt{6}\right)^2} \\ +\frac{\left(\frac{0.41669}{3}\right)(3)}{(s+3)^2-\left(\sqrt{6}\right)^2} \end{array}\right)$$

$$=\left(\begin{array}{l} 0.08333\cos 3t+0.0277\sin(3t)-0.08333\cosh\left(\sqrt{6}t\right) \\ +0.102\sinh\left(\sqrt{6}t\right)+0.13889\sinh\left(\sqrt{6}t\right) \end{array}\right)$$

Therefore, the inverse Laplace transform of $F(s)$ is

$$\boxed{0.08333\cos 3t+0.0277\sin(3t)-0.08333\cosh\left(\sqrt{6}t\right)+0.24089\sinh\left(\sqrt{6}t\right)}.$$